

AUTO-CAD 2D LAB MANUAL

Experiment 1: Detail & Assembly Drawing using AutoCAD (2D)

1. Objective

To create 2D detailed and assembly drawings of basic machine components using AutoCAD with proper dimensioning and A3 sheet layout.

2. Software & Tools

AutoCAD (2D)

Computer system

Standard drawing sheet (A3)

3. Drawing Standards (Simplified Academic)

Units: mm

Scale: 1:1

Projection: First Angle

Line types:

Object: Continuous thick

Center: Dashed

Dimension: Thin

4. A3 Sheet Layout

Sheet Size

420 × 297 mm

Margins

Left: 20 mm

Other sides: 10 mm

Title Block (Bottom Right)

Include:

Name

Roll No

Experiment No

Scale

Date

5. General AutoCAD Procedure

Open AutoCAD → New Drawing

Set Units (UNITS) → mm

Set Limits → 0,0 to 420,297

Create Layers (Object, Center, Dimension)

Draw using LINE, CIRCLE, ARC

Modify using OFFSET, TRIM, FILLET

Add Dimensions (DIM)

Add Hatch (for section)

Plot → A3 → Scale 1:1

6. COMPONENT DRAWINGS (WITH SIMPLE DIMENSIONS)

6.1 Plummer Block

Dimensions (mm)

Base: 120 × 80

Height: 80

Bearing hole: Ø40

Bolt holes: Ø10 (2 holes)

Bolt spacing: 80 mm

Steps

Draw base rectangle (120 × 80)

Draw centerline
Draw semicircle ($\varnothing 40$) on top
Add side walls
Draw bolt holes
Apply hatch in section view
Add dimensions
6.2 Wall Bracket
Dimensions
Base: 100×60
Vertical height: 120
Rib thickness: 10
Holes: $\varnothing 10$
Steps
Draw vertical plate
Draw horizontal base
Add inclined rib
Create holes
Dimension
6.3 Stepped Pulley
Dimensions
Outer diameters: $\varnothing 100$, $\varnothing 80$, $\varnothing 60$
Shaft hole: $\varnothing 20$
Width: 60
Steps
Draw centerline
Draw concentric circles
Offset for steps
Add keyway
Dimension diameters
6.4 Flanged Coupling
Dimensions
Flange diameter: $\varnothing 120$
Shaft hole: $\varnothing 30$
Bolt circle: $\varnothing 90$
Bolt holes: $\varnothing 10$ (4 holes)
Steps
Draw flange circle
Draw shaft hole
Create bolt circle
Use ARRAY for 4 bolts
Add section view
Dimension
6.5 Machine Tool Holder (3 Views)
Dimensions
Length: 100
Width: 40
Height: 60
Steps
Draw front view
Project top view
Project side view
Maintain alignment
Add dimensions

6.6 Screw Jack

Dimensions

Base: 80 × 80

Height: 150

Screw diameter: Ø25

Head: Ø50

Steps

Draw base

Draw vertical screw

Add head

Show section

Add dimensions

7. Sectional Views

Use HATCH

Angle: 45°

Apply only on cut surfaces

8. Dimensioning Rules

Do not repeat dimensions

Keep dimensions outside object

Use proper symbols (Ø for diameter)

9. Precautions

Maintain scale 1:1

Use correct layers

Keep drawing neat

Avoid overlapping lines

10. Result

All machine components were successfully drawn using AutoCAD (2D) with proper dimensioning and A3 layout.

11. Viva Questions (Important)

What is AutoCAD?

Difference between detail and assembly drawing

What is first angle projection?

What is the use of hatch?

Explain OFFSET and TRIM commands

What is a sectional view?